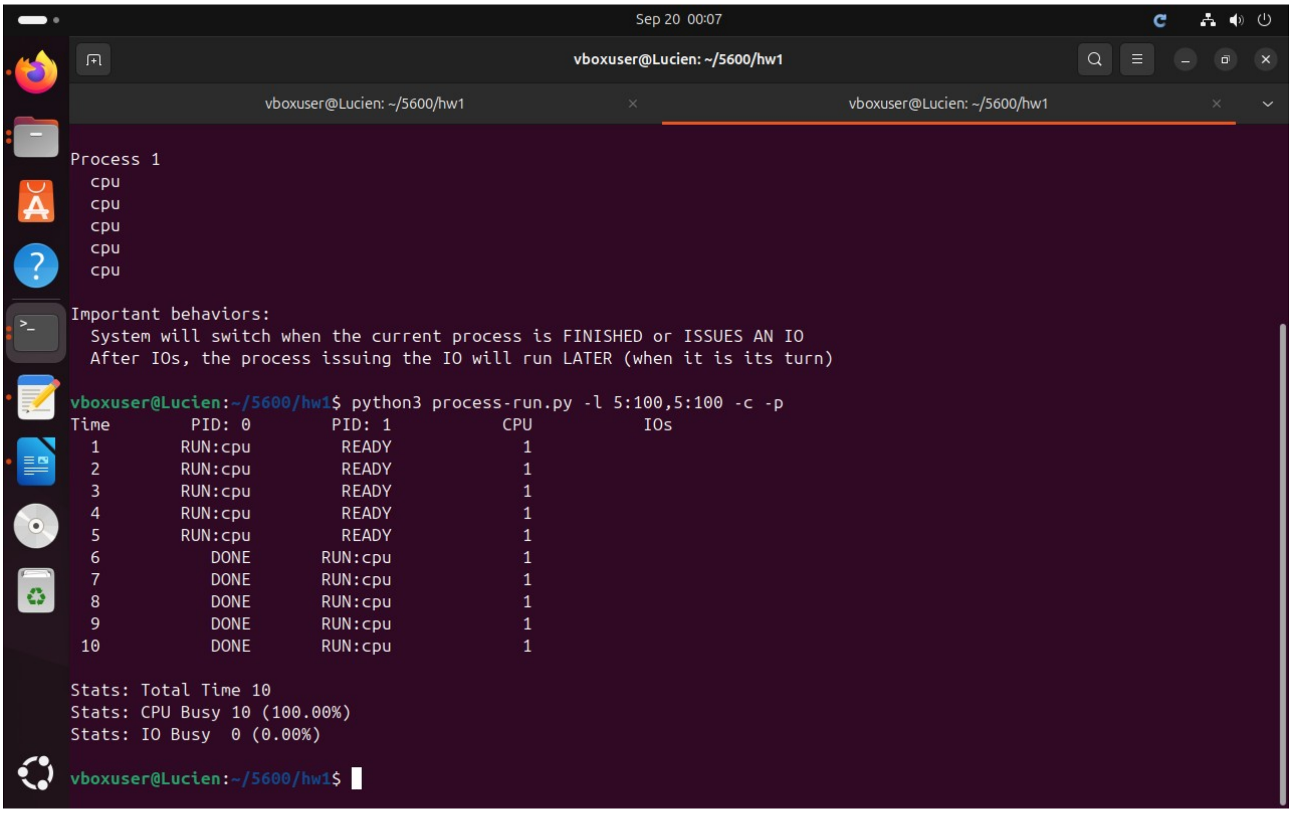
CS5600 HW 1
Lucien Roberson

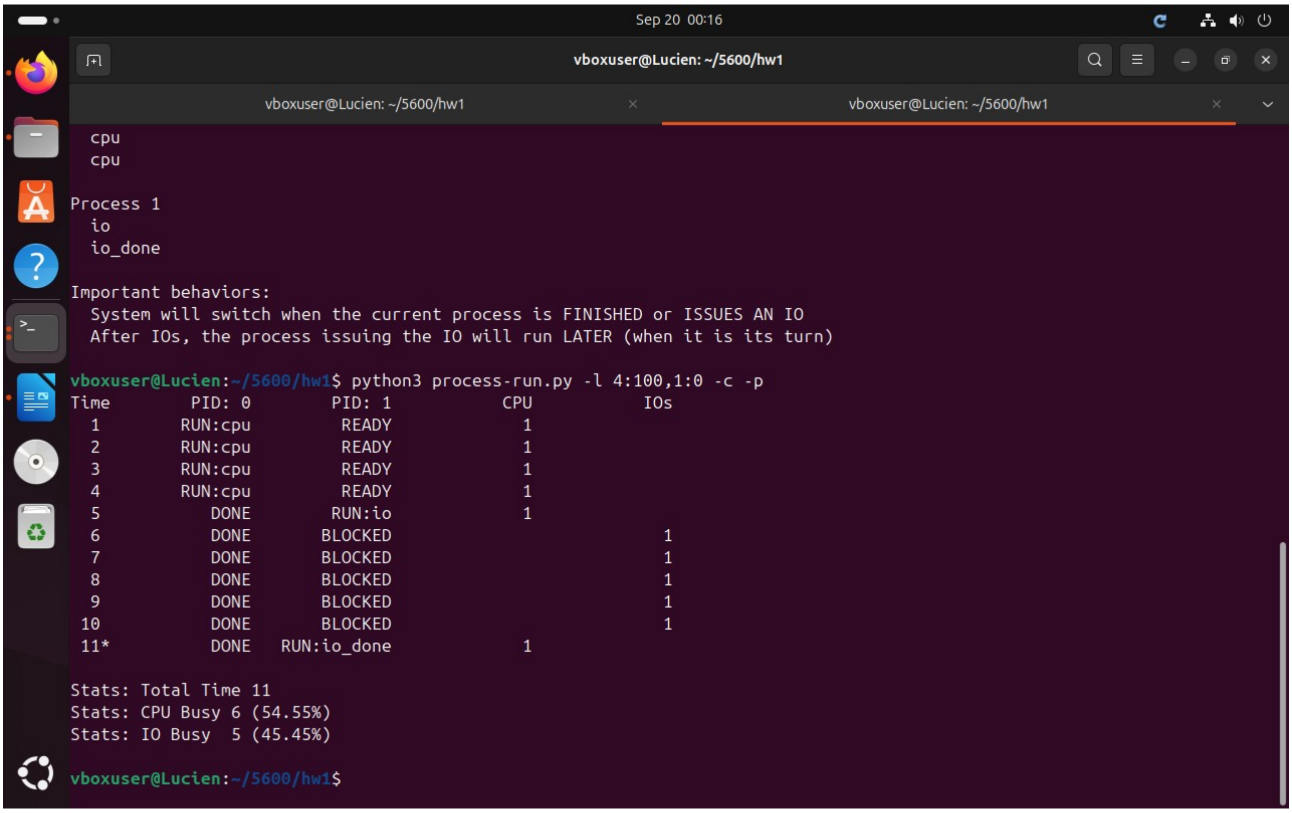
#1


Sep 20 00:07
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
Process 1
  cpu
  cpu
  cpu
  cpu
  cpu

Important behaviors:
  System will switch when the current process is FINISHED or ISSUES AN IO
  After IOs, the process issuing the IO will run LATER (when it is its turn)

vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 5:100,5:100 -c -p
Time        PID: 0        PID: 1           CPU           IOs
  1        RUN:cpu         READY             1
  2        RUN:cpu         READY             1
  3        RUN:cpu         READY             1
  4        RUN:cpu         READY             1
  5        RUN:cpu         READY             1
  6           DONE       RUN:cpu             1
  7           DONE       RUN:cpu             1
  8           DONE       RUN:cpu             1
  9           DONE       RUN:cpu             1
 10           DONE       RUN:cpu             1

Stats: Total Time 10
Stats: CPU Busy 10 (100.00%)
Stats: IO Busy  0 (0.00%)

vboxuser@Lucien:~/5600/hw1$ 
```

The :100 argument means it uses 100% of the CPU.

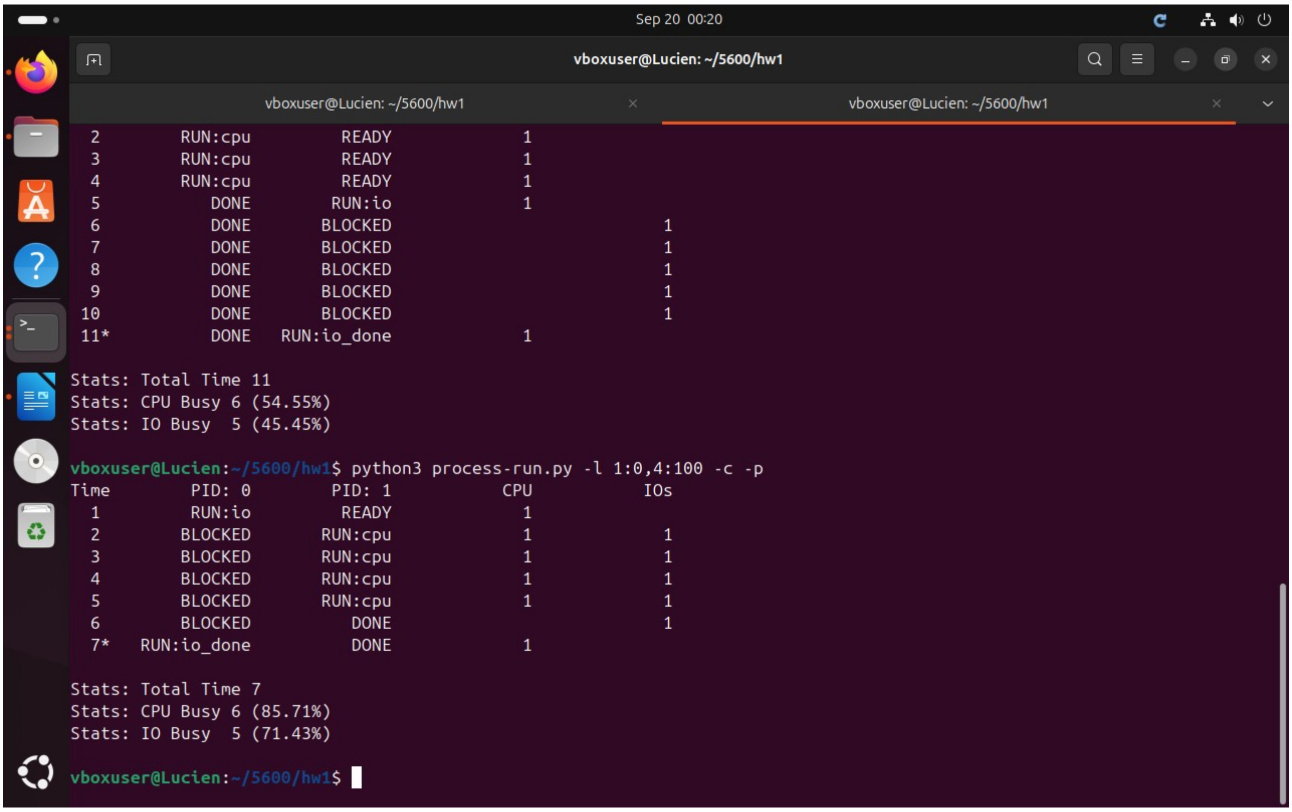
#2


Sep 20 00:16
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
  cpu
  cpu

Process 1
  io
  io_done

Important behaviors:
  System will switch when the current process is FINISHED or ISSUES AN IO
  After IOs, the process issuing the IO will run LATER (when it is its turn)

vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 4:100,1:0 -c -p
Time        PID: 0        PID: 1           CPU           IOs
  1        RUN:cpu         READY             1
  2        RUN:cpu         READY             1
  3        RUN:cpu         READY             1
  4        RUN:cpu         READY             1
  5           DONE        RUN:io             1
  6           DONE       BLOCKED                           1
  7           DONE       BLOCKED                           1
  8           DONE       BLOCKED                           1
  9           DONE       BLOCKED                           1
 10           DONE       BLOCKED                           1
 11*          DONE   RUN:io_done             1

Stats: Total Time 11
Stats: CPU Busy 6 (54.55%)
Stats: IO Busy  5 (45.45%)

vboxuser@Lucien:~/5600/hw1$
```

It takes 11 total steps, 1 per CPU op, 5 per io op, 1 per io start and stop

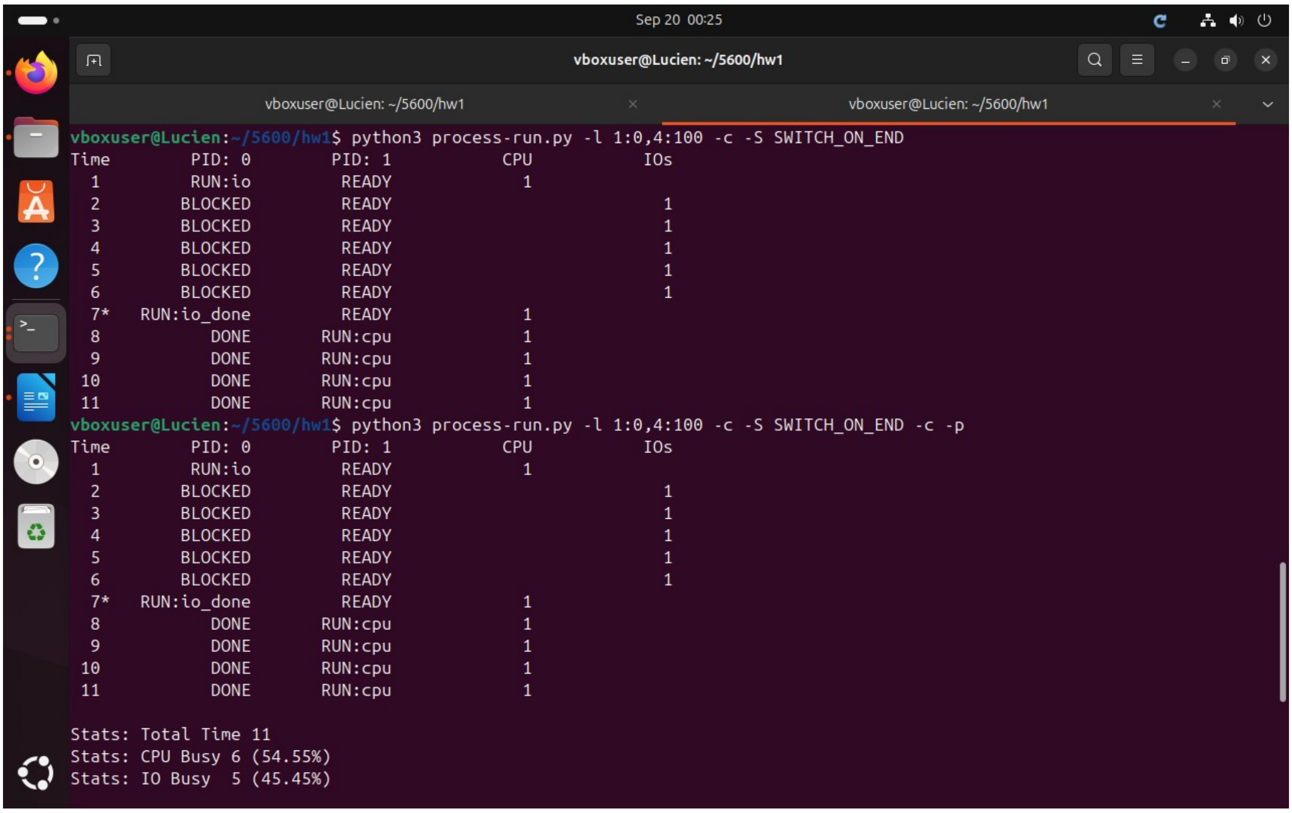
#3


Sep 20 00:20
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
  2        RUN:cpu         READY             1
  3        RUN:cpu         READY             1
  4        RUN:cpu         READY             1
  5           DONE        RUN:io             1
  6           DONE       BLOCKED                           1
  7           DONE       BLOCKED                           1
  8           DONE       BLOCKED                           1
  9           DONE       BLOCKED                           1
 10           DONE       BLOCKED                           1
 11*          DONE   RUN:io_done             1

Stats: Total Time 11
Stats: CPU Busy 6 (54.55%)
Stats: IO Busy  5 (45.45%)

vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 1:0,4:100 -c -p
Time        PID: 0        PID: 1           CPU           IOs
  1         RUN:io         READY             1
  2        BLOCKED       RUN:cpu             1             1
  3        BLOCKED       RUN:cpu             1             1
  4        BLOCKED       RUN:cpu             1             1
  5        BLOCKED       RUN:cpu             1             1
  6        BLOCKED          DONE                           1
  7*   RUN:io_done          DONE             1

Stats: Total Time 7
Stats: CPU Busy 6 (85.71%)
Stats: IO Busy  5 (71.43%)

vboxuser@Lucien:~/5600/hw1$
```

1 per io start and stop, 1 per cpu op, 1 per io op.

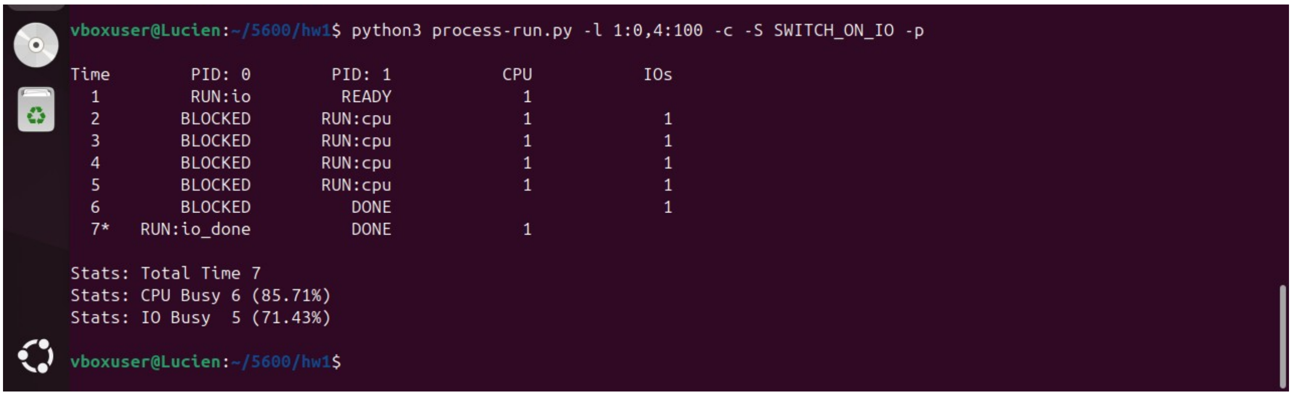
#4


Sep 20 00:25
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 1:0,4:100 -c -S SWITCH_ON_END
Time        PID: 0        PID: 1           CPU           IOs
  1         RUN:io         READY             1
  2        BLOCKED         READY                           1
  3        BLOCKED         READY                           1
  4        BLOCKED         READY                           1
  5        BLOCKED         READY                           1
  6        BLOCKED         READY                           1
  7*   RUN:io_done         READY             1
  8           DONE       RUN:cpu             1
  9           DONE       RUN:cpu             1
 10           DONE       RUN:cpu             1
 11           DONE       RUN:cpu             1
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 1:0,4:100 -c -S SWITCH_ON_END -c -p
Time        PID: 0        PID: 1           CPU           IOs
  1         RUN:io         READY             1
  2        BLOCKED         READY                           1
  3        BLOCKED         READY                           1
  4        BLOCKED         READY                           1
  5        BLOCKED         READY                           1
  6        BLOCKED         READY                           1
  7*   RUN:io_done         READY             1
  8           DONE       RUN:cpu             1
  9           DONE       RUN:cpu             1
 10           DONE       RUN:cpu             1
 11           DONE       RUN:cpu             1

Stats: Total Time 11
Stats: CPU Busy 6 (54.55%)
Stats: IO Busy  5 (45.45%)
```

It now takes 11 steps as CPU is now blocked until the end of the IO ops

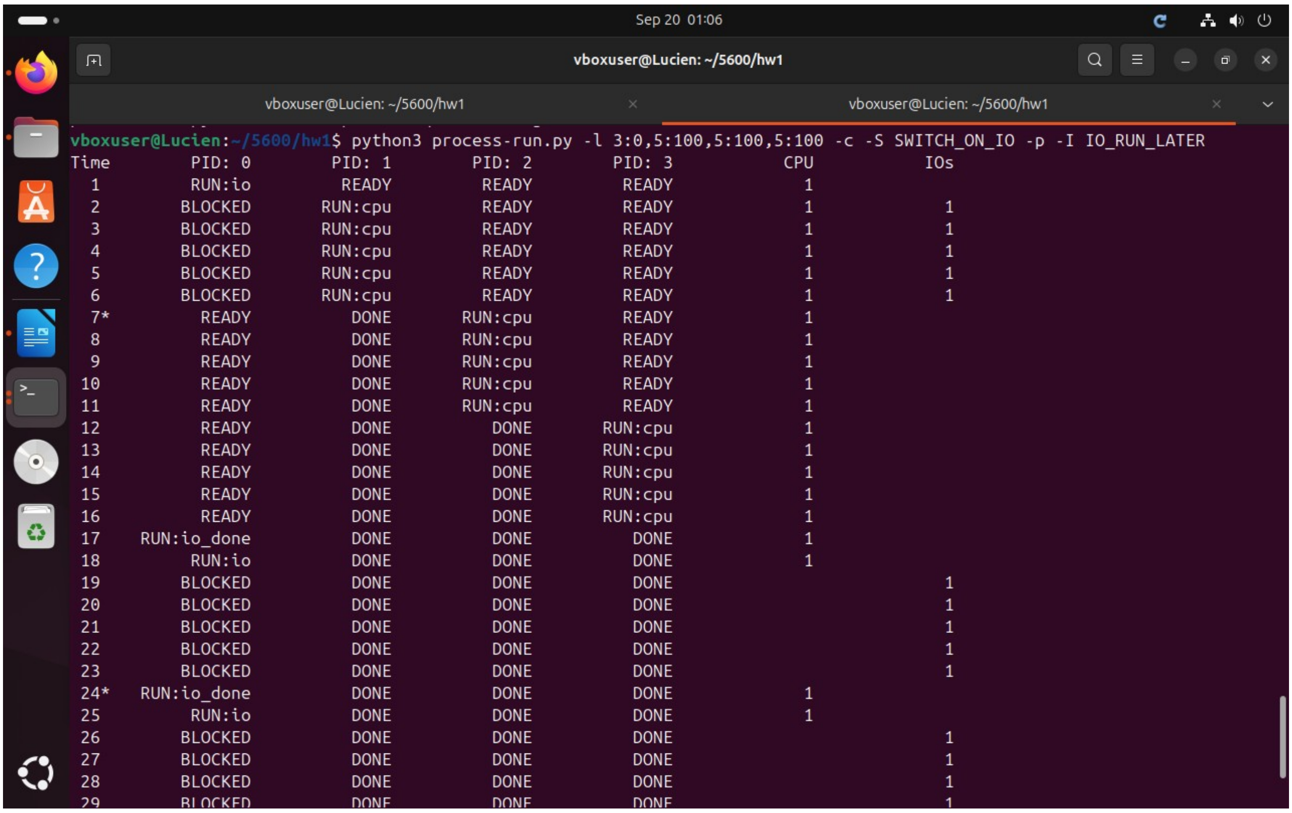
#5



```
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 1:0,4:100 -c -S SWITCH_ON_IO -p

Time        PID: 0        PID: 1           CPU           IOs
  1         RUN:io         READY             1
  2        BLOCKED       RUN:cpu             1             1
  3        BLOCKED       RUN:cpu             1             1
  4        BLOCKED       RUN:cpu             1             1
  5        BLOCKED       RUN:cpu             1             1
  6        BLOCKED          DONE                           1
  7*   RUN:io_done          DONE             1

Stats: Total Time 7
Stats: CPU Busy 6 (85.71%)
Stats: IO Busy  5 (71.43%)

vboxuser@Lucien:~/5600/hw1$
```

CPU process is not waiting IO to end

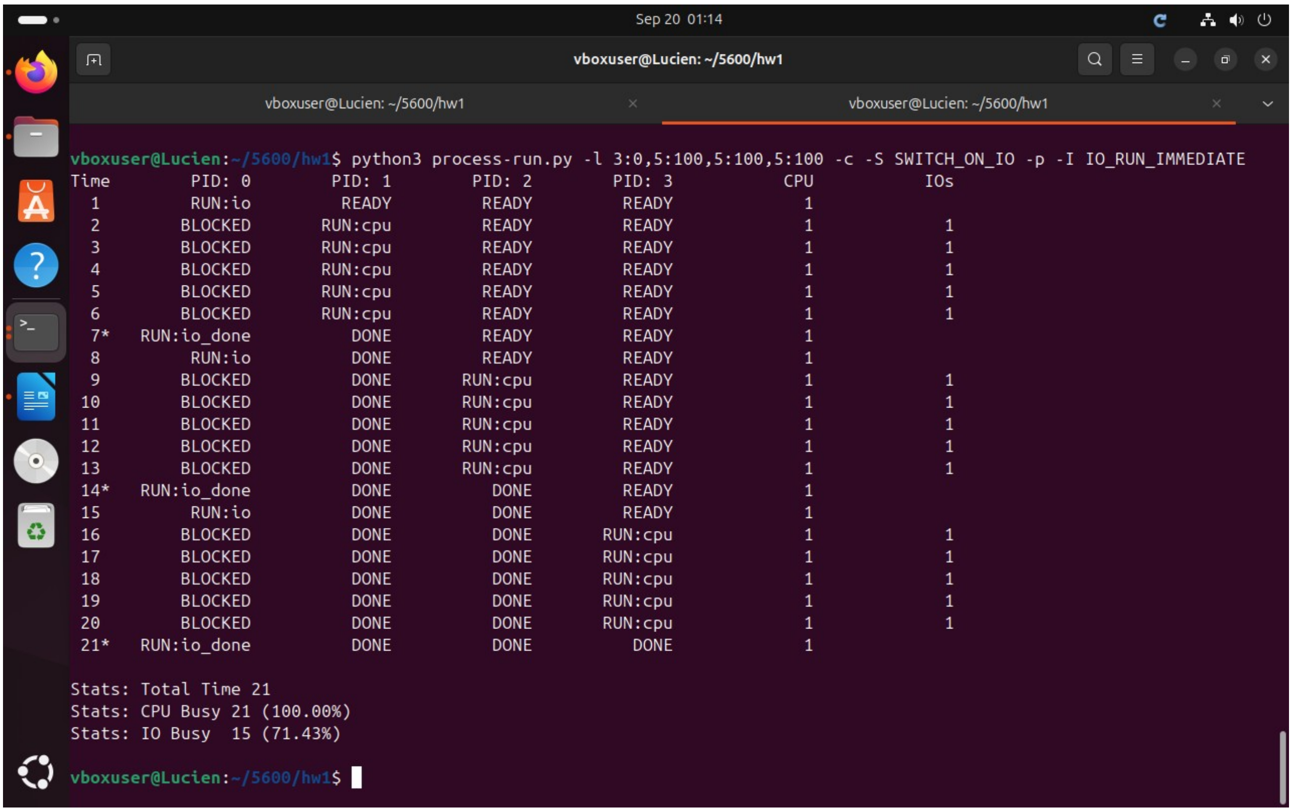
#6


Sep 20 01:06
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 3:0,5:100,5:100,5:100 -c -S SWITCH_ON_IO -p -I IO_RUN_LATER
Time        PID: 0       PID: 1       PID: 2       PID: 3          CPU           IOs
  1         RUN:io        READY        READY        READY            1
  2        BLOCKED      RUN:cpu        READY        READY            1             1
  3        BLOCKED      RUN:cpu        READY        READY            1             1
  4        BLOCKED      RUN:cpu        READY        READY            1             1
  5        BLOCKED      RUN:cpu        READY        READY            1             1
  6        BLOCKED      RUN:cpu        READY        READY            1             1
  7*         READY         DONE      RUN:cpu        READY            1
  8          READY         DONE      RUN:cpu        READY            1
  9          READY         DONE      RUN:cpu        READY            1
 10          READY         DONE      RUN:cpu        READY            1
 11          READY         DONE      RUN:cpu        READY            1
 12          READY         DONE         DONE      RUN:cpu            1
 13          READY         DONE         DONE      RUN:cpu            1
 14          READY         DONE         DONE      RUN:cpu            1
 15          READY         DONE         DONE      RUN:cpu            1
 16          READY         DONE         DONE      RUN:cpu            1
 17    RUN:io_done         DONE         DONE         DONE            1
 18         RUN:io         DONE         DONE         DONE            1
 19        BLOCKED         DONE         DONE         DONE                          1
 20        BLOCKED         DONE         DONE         DONE                          1
 21        BLOCKED         DONE         DONE         DONE                          1
 22        BLOCKED         DONE         DONE         DONE                          1
 23        BLOCKED         DONE         DONE         DONE                          1
 24*   RUN:io_done         DONE         DONE         DONE            1
 25         RUN:io         DONE         DONE         DONE            1
 26        BLOCKED         DONE         DONE         DONE                          1
 27        BLOCKED         DONE         DONE         DONE                          1
 28        BLOCKED         DONE         DONE         DONE                          1
 29        BLOCKED         DONE         DONE         DONE                          1
```

System resources are not being used efficiently, IO waits unnecessarily after P0 finishes

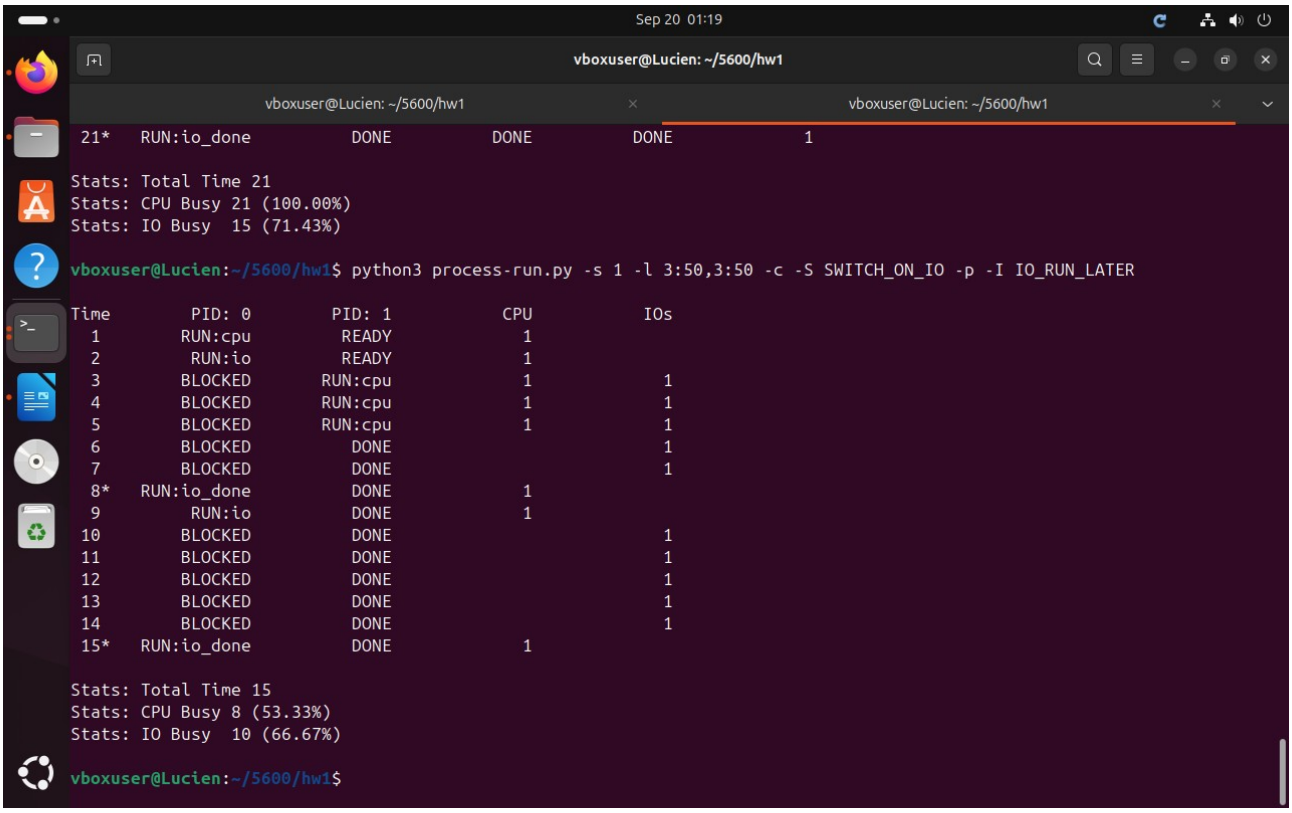
#7


Sep 20 01:14
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

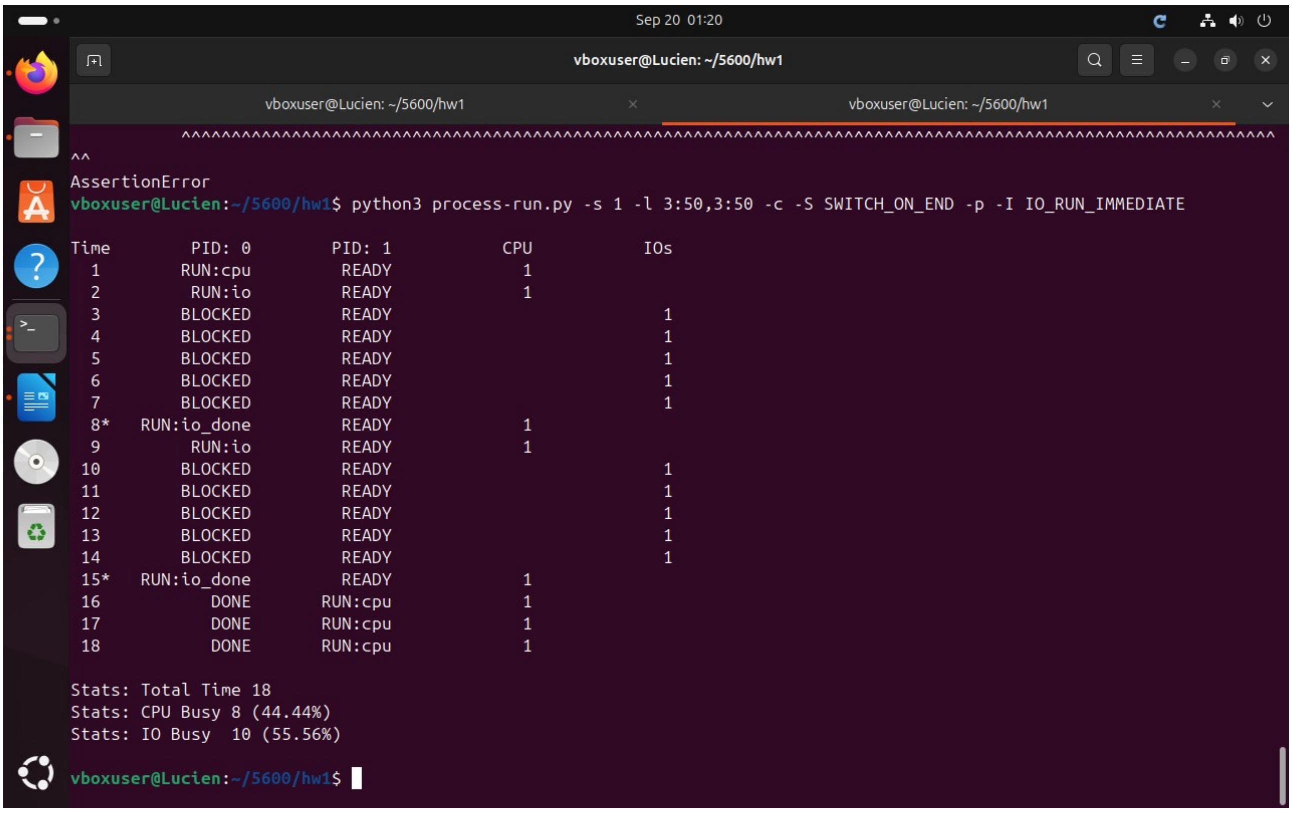
```
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -l 3:0,5:100,5:100,5:100 -c -S SWITCH_ON_IO -p -I IO_RUN_IMMEDIATE
Time        PID: 0       PID: 1       PID: 2       PID: 3          CPU           IOs
  1         RUN:io        READY        READY        READY            1
  2        BLOCKED      RUN:cpu        READY        READY            1             1
  3        BLOCKED      RUN:cpu        READY        READY            1             1
  4        BLOCKED      RUN:cpu        READY        READY            1             1
  5        BLOCKED      RUN:cpu        READY        READY            1             1
  6        BLOCKED      RUN:cpu        READY        READY            1             1
  7*   RUN:io_done         DONE        READY        READY            1
  8         RUN:io         DONE        READY        READY            1
  9        BLOCKED         DONE      RUN:cpu        READY            1             1
 10        BLOCKED         DONE      RUN:cpu        READY            1             1
 11        BLOCKED         DONE      RUN:cpu        READY            1             1
 12        BLOCKED         DONE      RUN:cpu        READY            1             1
 13        BLOCKED         DONE      RUN:cpu        READY            1             1
 14*   RUN:io_done         DONE         DONE        READY            1
 15         RUN:io         DONE         DONE        READY            1
 16        BLOCKED         DONE         DONE      RUN:cpu            1             1
 17        BLOCKED         DONE         DONE      RUN:cpu            1             1
 18        BLOCKED         DONE         DONE      RUN:cpu            1             1
 19        BLOCKED         DONE         DONE      RUN:cpu            1             1
 20        BLOCKED         DONE         DONE      RUN:cpu            1             1
 21*   RUN:io_done         DONE         DONE         DONE            1

Stats: Total Time 21
Stats: CPU Busy 21 (100.00%)
Stats: IO Busy  15 (71.43%)

vboxuser@Lucien:~/5600/hw1$
```

It is now running efficiently as processes do not have to wait on the CPU

#8


Sep 20 01:19
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
 21*   RUN:io_done          DONE          DONE          DONE             1

Stats: Total Time 21
Stats: CPU Busy 21 (100.00%)
Stats: IO Busy  15 (71.43%)

vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -s 1 -l 3:50,3:50 -c -S SWITCH_ON_IO -p -I IO_RUN_LATER

Time        PID: 0        PID: 1           CPU           IOs
  1        RUN:cpu         READY             1
  2         RUN:io         READY             1
  3        BLOCKED       RUN:cpu             1             1
  4        BLOCKED       RUN:cpu             1             1
  5        BLOCKED       RUN:cpu             1             1
  6        BLOCKED          DONE                           1
  7        BLOCKED          DONE                           1
  8*   RUN:io_done          DONE             1
  9         RUN:io          DONE             1
 10        BLOCKED          DONE                           1
 11        BLOCKED          DONE                           1
 12        BLOCKED          DONE                           1
 13        BLOCKED          DONE                           1
 14        BLOCKED          DONE                           1
 15*   RUN:io_done          DONE             1

Stats: Total Time 15
Stats: CPU Busy 8 (53.33%)
Stats: IO Busy  10 (66.67%)

vboxuser@Lucien:~/5600/hw1$
```


Sep 20 01:20
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1
vboxuser@Lucien: ~/5600/hw1

```
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^
AssertionError
vboxuser@Lucien:~/5600/hw1$ python3 process-run.py -s 1 -l 3:50,3:50 -c -S SWITCH_ON_END -p -I IO_RUN_IMMEDIATE

Time        PID: 0        PID: 1           CPU           IOs
  1        RUN:cpu         READY             1
  2         RUN:io         READY             1
  3        BLOCKED         READY                           1
  4        BLOCKED         READY                           1
  5        BLOCKED         READY                           1
  6        BLOCKED         READY                           1
  7        BLOCKED         READY                           1
  8*   RUN:io_done         READY             1
  9         RUN:io         READY             1
 10        BLOCKED         READY                           1
 11        BLOCKED         READY                           1
 12        BLOCKED         READY                           1
 13        BLOCKED         READY                           1
 14        BLOCKED         READY                           1
 15*   RUN:io_done         READY             1
 16           DONE       RUN:cpu             1
 17           DONE       RUN:cpu             1
 18           DONE       RUN:cpu             1

Stats: Total Time 18
Stats: CPU Busy 8 (44.44%)
Stats: IO Busy  10 (55.56%)

vboxuser@Lucien:~/5600/hw1$
```

SWITCH_ON_IO made a bigger difference than IO_RUN_IMMEDIATE